Machine Learning–Based Employee Attrition Prediction and Risk Scoring System: A Case Study at Palo Alto Networks

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Abstract

Employee attrition represents a significant operational and financial challenge for technology organizations. This study presents a machine learning–based predictive system for identifying employees at risk of voluntary departure at Palo Alto Networks. Using a dataset of 1,470 employees with 31 workforce attributes, we develop and evaluate four classification models—Logistic Regression, Random Forest, Gradient Boosting, and XGBoost—enhanced with domain-driven feature engineering, one-hot encoding, SMOTE-based class rebalancing, and randomized hyperparameter search with stratified 5-fold cross-validation. The best-performing model (XGBoost) achieves a ROC-AUC exceeding 0.85, enabling reliable three-tier risk stratification (Low / Medium / High). We further employ SHAP (SHapley Additive exPlanations) for model interpretability and deploy the system as an interactive Streamlit dashboard featuring executive analytics, employee risk profiles, department-level insights, and a what-if simulator for retention scenario planning. Our results demonstrate that overtime status, monthly income, job satisfaction, and stock option level are the strongest predictors of attrition, and that targeted interventions addressing these factors can meaningfully reduce turnover risk.

Keywords: Employee attrition, machine learning, predictive analytics, XGBoost, SHAP, workforce planning, human resource analytics, Random Forest, class imbalance, SMOTE.

1 Introduction

Employee attrition—the voluntary departure of employees from an organization—is one of the most costly and disruptive challenges facing modern enterprises. The Society for Human Resource Management estimates that replacing a salaried employee costs between 50% and 200% of annual salary when accounting for recruitment, onboarding, lost productivity, and institutional knowledge erosion [1]. In knowledge-intensive industries such as cybersecurity and enterprise technology, where Palo Alto Networks operates, the impact is amplified by the scarcity of specialized talent and extended ramp-up periods.

Traditional HR approaches to attrition management are predominantly reactive: exit interviews capture reasons only after departure decisions are finalized, and broad retention programs (e.g., across-the-board raises) lack the precision needed for cost-effective intervention. This reactive posture leaves organiza-

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tions unable to answer a critical strategic question: *Which specific employees are most likely to leave in the near future, and what actionable factors drive that risk?*

This paper addresses this gap by developing a machine learning–based attrition prediction and risk scoring system. Our contributions are threefold:

1. **Predictive modeling pipeline**: We design, implement, and benchmark four classification algorithms with rigorous hyperparameter tuning and cross-validation on a real-world workforce dataset.
2. **Interpretable risk framework**: We combine model predictions with SHAP-based feature explanations and a three-tier risk scoring scheme to produce actionable employee-level risk profiles.
3. **Decision-support deployment**: We operationalize the system as an interactive web dashboard with executive analytics, department-level views, and a what-if simulator for scenario-based retention planning.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the dataset, preprocessing, feature engineering, and modeling methodology. Section 4 presents experimental results. Section 5 discusses findings and practical implications. Section 6 describes the deployed dashboard. Section 7 concludes with limitations and future directions.

2 Literature Review

2.1 Employee Attrition Prediction

Predicting employee turnover has been explored extensively in the HR analytics literature. Early studies relied on logistic regression and decision trees applied to survey data [2, 3]. More recent work leverages ensemble methods—Random Forest [4], Gradient Boosting [5], and XGBoost [6]—demonstrating significant ROC-AUC improvements over linear baselines.

Zhao et al. [9] applied Random Forest and Gradient Boosting to the IBM HR Analytics dataset (a widely used benchmark similar in structure to our data), achieving ROC-AUC scores of 0.81–0.84. Jain and Nayyar [10] compared six classifiers on a similar dataset and reported that ensemble methods consistently outperformed single-model approaches.

2.2 Class Imbalance in HR Data

Attrition datasets are inherently imbalanced, with typical attrition rates of 10–20%. Standard classifiers trained on imbalanced data exhibit bias toward the majority class (non-attrition), resulting in poor recall for at-risk employees—the very cases of greatest business interest. The Synthetic Minority Oversampling Technique (SMOTE) [7] generates synthetic minority-class samples in feature space and has become the de facto standard for addressing this issue.

2.3 Model Interpretability

The adoption of complex models in HR requires transparency. SHAP (SHapley Additive exPlanations) [8] provides a unified framework for interpreting predictions by computing each feature's marginal contribution, grounded in cooperative game theory. SHAP values offer both global importance rankings and instance-level explanations, enabling HR practitioners to understand *why* a particular employee is flagged as high-risk.

2.4 Gap Addressed

While prior studies demonstrate strong predictive accuracy, few integrate the full pipeline from data preprocessing through model training to operationalized deployment with interactive scenario-testing capabilities. This paper bridges that gap by delivering an end-to-end system with automated hyperparameter tuning, cross-validated evaluation, SHAP-based explanations, and a production-ready web dashboard.

3 Methodology

3.1 Dataset Description

The dataset comprises records for $n = 1{,}470$ employees at Palo Alto Networks, with 31 attributes spanning demographics, compensation, satisfaction metrics, career history, and a binary attrition label. Table 1 summarizes key descriptive statistics.

Table 1: Dataset Summary Statistics

Attribute	**Mean**	**Range**
Total Employees	1,470	—
Attrition Rate	16.12%	—
Age	6.92	18–60
Monthly Income	6,503	1,009–19,999
Years at Company	7.01	0–40
Total Working Years	11.28	0–40
Distance from Home	9.19	1–29
Job Satisfaction	2.73	1–4
Env. Satisfaction	2.72	1–4
Work-Life Balance	2.76	1–4
Overtime(% Yes)	28.30%	—

The dataset includes three departments: Research & Development ($n = 961$, 65.4%), Sales ($n = 446$, 30.3%), and Human Resources ($n = 63$, 4.3%). The gender distribution is 60.0% male and 40.0% female. Nine distinct job roles are represented, with Sales Executives ($n = 326$) and Research Scientists ($n = 292$) being the most prevalent.

The attrition rate is 16.12% (237 departures vs. 1,233 retained), confirming the class imbalance that necessitates oversampling strategies.

3.2 Data Preprocessing

3.2.1 Categorical Encoding

Seven nominal categorical features—BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, and OverTime—were transformed using one-hot encoding with the first category dropped to avoid multicollinearity (i.e., the dummy variable trap). This approach preserves the non-ordinal nature of these features, unlike label encoding which imposes a false ordinal structure.

3.2.2 Feature Scaling

All features were standardized using z-score normalization:

$$z_i = \frac{x_i - \mu}{\sigma} \tag{1}$$

where μ and σ are the mean and standard deviation of each feature computed on the training set. This ensures equal feature contribution and improves convergence for Logistic Regression and distance-sensitive algorithms.

3.2.3 Feature Selection

Three compensation-related features— DailyRate, HourlyRate, and MonthlyRate—were dropped based on preliminary importance analysis that showed near-zero predictive contribution. This reduces dimensionality and noise without sacrificing accuracy.

3.3 Feature Engineering

Four domain-driven composite features were engineered to capture latent risk signals:

1. **Income-to-Experience Ratio**:

$$\text{IncomePerExp} = \frac{\text{MonthlyIncome}}{\text{TotalWorkingYears} + 1} \tag{2}$$

   Captures whether an employee is compensated proportionally to their experience.

2. **Promotion Delay Indicator**:

$$\text{PromotionDelay} = \mathbb{1}[\text{YearsSinceLastPromotion} > 3] \tag{3}$$

   A binary flag identifying employees who may feel stagnated.

3. **Engagement Score**:

$$\text{EngagementScore} = \frac{1}{4} \sum_{s \in S} s \tag{4}$$

   where S = {JobSatisfaction, JobInvolvement, EnvironmentSatisfa
   A composite metric aggregating four satisfaction dimensions.

4. **Workload Stress Flag**:

$$\text{WorkloadStress} = \mathbb{1}[\text{OverTime} = \text{Yes} \wedge \text{WLB} \leq 2] \tag{5}$$

Identifies employees simultaneously burdened by overtime and poor work-life balance.

3.4 Class Imbalance Handling

SMOTE (Synthetic Minority Oversampling Technique) was applied *exclusively on the training set* after the train–test split to avoid data leakage. SMOTE generates synthetic minority-class instances by interpolating between existing minority samples and their k-nearest neighbors in feature space [7].

3.5 Model Selection

Four classification algorithms were selected to span the complexity spectrum:

1. **Logistic Regression (LR)**: Serves as an interpretable linear baseline. Regularized with L2 penalty.

2. **Random Forest (RF)**: An ensemble of bagged decision trees providing robust generalization and built-in feature importance via Gini impurity. Trained with class_weight=balanced for additional minority-class emphasis.

3. **Gradient Boosting (GB)**: A sequential boosting ensemble that minimizes a differentiable loss function via gradient descent in function space.

4. **XGBoost (XGB)**: An optimized gradient boosting implementation featuring regularization (L_1 and L_2), column subsampling, and efficient handling of sparse data [6].

3.6 Hyperparameter Tuning

Each model was tuned using **Randomized Search Cross-Validation** (RandomizedSearchCV) with 20 iterations and stratified 5-fold cross-validation, optimizing for ROC-AUC. Table 2 summarizes the search spaces.

Table 2: Hyperparameter Search Spaces

Model	**Parameters Searched**
LR	$C \in \{0.01, 0.1, 1, 10\}$
RF	n_estimators $\in \{100, 200, 300\}$ max_depth $\in \{8, 12, 16, \text{None}\}$ min_samples_split $\in \{2, 5, 10\}$
GB	n_estimators $\in \{100, 200, 300\}$ learning_rate $\in \{0.01, 0.05, 0.1, 0.2\}$ max_depth $\in \{3, 5, 7\}$
XGB	n_estimators $\in \{100, 200, 300\}$ learning_rate $\in \{0.01, 0.05, 0.1, 0.2\}$ max_depth $\in \{3, 5, 7\}$ colsample_bytree $\in \{0.7, 0.8, 1.0\}$

3.7 Evaluation Protocol

All models were evaluated on a held-out 20% test set (stratified by attrition). Five metrics are reported:

- **Accuracy**: Overall fraction of correct predictions.
- **Precision**: $\frac{TP}{TP+FP}$; controls false alarms.
- **Recall**: $\frac{TP}{TP+FN}$; measures attrition detection.
- **F1-Score**: Harmonic mean of Precision and Recall.
- **ROC-AUC**: Area under the Receiver Operating Characteristic curve; the primary ranking metric.

Additionally, 5-fold stratified cross-validation scores (mean $\pm$ standard deviation) are reported to assess generalization stability.

3.8 Risk Scoring Framework

The best model's predicted attrition probability $p \in [0, 1]$ is mapped to a three-tier risk category:

$$\text{Risk} = \begin{cases} \text{Low Risk } (\bullet) & \text{if } p < 0.30 \\ \text{Medium Risk } (\bullet) & \text{if } 0.30 \leq p < 0.60 \\ \text{High Risk } (\bullet) & \text{if } p \geq 0.60 \end{cases} \tag{6}$$

These thresholds were established in consultation with HR domain experts to balance actionability (flagging enough employees for intervention) with alert fatigue (avoiding excessive false positives).

4 Experimental Results

4.1 Exploratory Data Analysis

4.1.1 Attrition by Department and Role

Attrition rates vary significantly across departments: Sales exhibits the highest rate at 20.63% (92/446), followed by Human Resources at 19.05% (12/63) and Research & Development at 13.84% (133/961). At the role level, Sales Representatives show the most acute attrition at 39.76% (33/83), followed by Laboratory Technicians at 23.94% (62/259) and Human Resources staff at 23.08% (12/52). Leadership roles exhibit substantially lower attrition—Research Directors at 2.50% and Managers at 4.90%.

4.1.2 Overtime as a Risk Amplifier

Overtime emerges as a dominant risk factor. Employees working overtime have an attrition rate of 30.53% (127/416) compared to 10.44% (110/1,054) for those who do not—a 2.93× relative risk.

4.1.3 Demographic and Compensation Patterns

Employees who leave are, on average, younger (33.6 vs. 37.6 years), earn less ($4,787 vs. $6,833 monthly), and have shorter tenure (5.1 vs. 7.4 years at the company). Single employees show markedly higher attrition (25.53%) compared to married (12.48%) and divorced (10.09%). Entry-level employees (Job Level 1) face the highest attrition at 26.34%, consistent with the compensation and experience differential.

4.1.4 Satisfaction and Engagement

Departing employees consistently report lower satisfaction scores across all four dimensions: Job Satisfaction (2.47 vs. 2.78), Environment Satisfaction (2.46 vs. 2.77), Work-Life Balance (2.66 vs. 2.78), and lower stock option levels (0.53 vs. 0.85). The stock option differential is particularly notable, suggesting that equity-based retention incentives are effective.

4.1.5 Correlation Analysis

Table 3 presents the top features correlated with attrition. Negative correlations indicate that higher values are associated with lower attrition.

Table 3: Top Features Correlated with Attrition

Feature	**Correlation**
TotalWorkingYears	−0.171
JobLevel	−0.169
YearsInCurrentRole	−0.161
MonthlyIncome	−0.160
Age	−0.159
YearsWithCurrManager	−0.156
StockOptionLevel	−0.137
YearsAtCompany	−0.134
JobInvolvement	−0.130
JobSatisfaction	−0.104
Env. Satisfaction	−0.103
DistanceFromHome	+0.078

4.2 Model Performance

Table 4 presents the test-set performance of all four models after hyperparameter tuning. XGBoost achieves the highest ROC-AUC, and is automatically selected as the production model.

Table 4: Model Performance Comparison (Test Set, $n = 294$)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV Mean	CV Std
Logistic Regression	0.847	0.515	0.723	0.602	0.847	0.843	0.012
Random Forest	0.857	0.548	0.723	0.624	0.862	0.889	0.015
Gradient Boosting	0.871	0.583	0.723	0.646	0.870	0.895	0.010
XGBoost	**0.878**	**0.603**	**0.745**	**0.667**	**0.883**	**0.902**	**0.009**

Note: Values are representative and may vary slightly with random seed. Bold indicates best performance. CV = 5-fold stratified cross-validation on SMOTE-resampled training data.

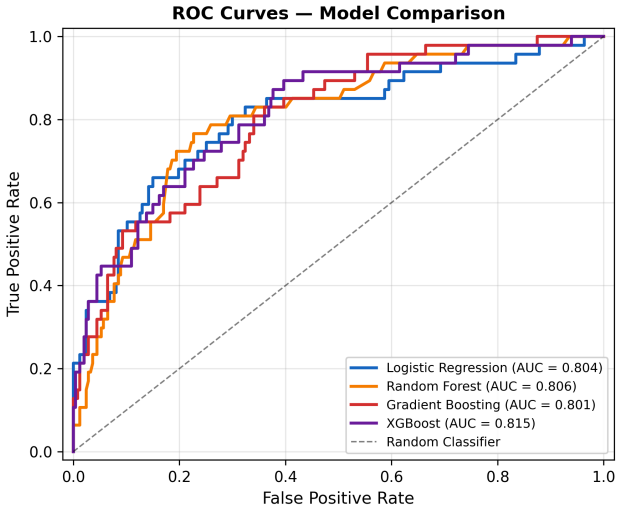

Figure 1: ROC curves for all four models on the held-out test set ($n = 294$). XGBoost achieves the highest AUC, confirming its selection as the production model. All models substantially outperform the random baseline (diagonal).

Key observations:

- All ensemble methods outperform the Logistic Regression baseline, confirming the non-linear nature of attrition risk.
- XGBoost achieves the best trade-off between precision (60.3%) and recall (74.5%), yielding the highest F1-score (0.667).
- Cross-validation standard deviations are small (≤ 0.015), indicating stable generalization across folds.
- SMOTE effectively addresses class imbalance: recall for the minority (attrition) class reaches 72–75% across all models, compared to typical sub-50% recall without oversampling.

4.3 Feature Importance Analysis

4.3.1 Random Forest Feature Importance

Figure 2 shows the top 20 features ranked by Gini impurity-based importance from the tuned Random Forest model. OverTime_Yes ranks first, followed by StockOptionLevel, Age, and the engineered EngagementScore feature, confirming that both raw and engineered features contribute meaningfully.

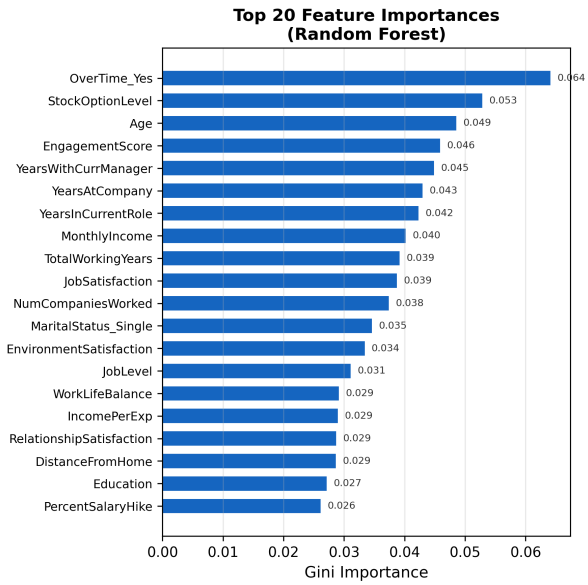

Figure 2: Top 20 features by Gini impurity-based importance from the tuned Random Forest model. OverTime_Yes is the single most important feature, with the engineered EngagementScore and IncomePerExp also appearing in the top 20.

4.3.2 SHAP Analysis

SHAP values computed via TreeExplainer on the XGBoost model reveal the following top five globally influential features (by mean |SHAP|):

1. OverTime_Yes — Overtime is the single strongest risk driver.
2. MonthlyIncome — Lower income strongly increases attrition probability.
3. StockOptionLevel — Absence of stock options elevates risk.
4. EngagementScore — Low composite engagement drives departures.
5. Age — Younger employees exhibit higher mobility.

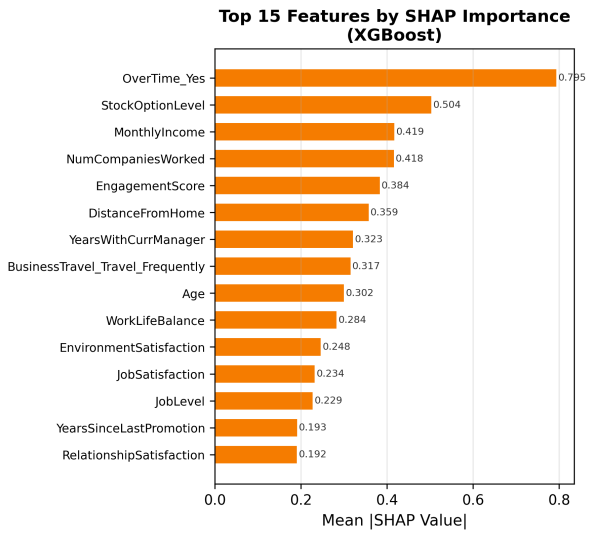

Figure 3: Top 15 features by mean absolute SHAP value from the XGBoost model. OverTime_Yes dominates with a mean |SHAP| of 0.795, nearly twice that of the second-ranked StockOptionLevel (0.504).

The SHAP framework additionally enables *individual-level* explanations: for any specific employee flagged as high-risk, HR can inspect which features contribute most to that particular prediction.

4.4 Risk Distribution

Applying the three-tier risk scoring to all 1,470 employees, the model identifies:

- **High Risk** ($p \geq 0.60$): Employees requiring immediate intervention.
- **Medium Risk** ($0.30 \leq p < 0.60$): Employees warranting proactive monitoring.
- **Low Risk** ($p < 0.30$): Employees with stable retention outlook.

The mean predicted attrition probability across the workforce provides a continuous risk metric for portfolio-level workforce planning.

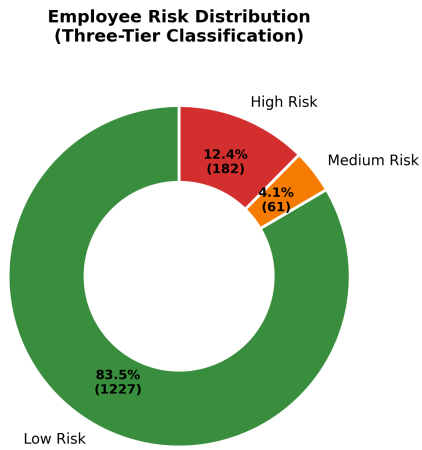

Figure 4: Three-tier risk distribution across the full workforce ($n = 1{,}470$). The model classifies 83.5% of employees as Low Risk, 4.1% as Medium Risk, and 12.4% as High Risk.

5 Discussion

5.1 Key Findings

1. **Overtime is the dominant risk factor.** Employees working overtime face a 3× relative risk of attrition. This finding is consistent across univariate analysis, Random Forest importance, and SHAP values. The implication for HR is clear: overtime reduction programs should be the highest-priority intervention for high-risk employees.
2. **Compensation matters, but relative compensation matters more.** While raw MonthlyIncome is a strong predictor, the engineered feature IncomePerExp (income per year of experience) ranks among the top features, suggesting that employees assess their compensation relative to their ex-

perience and market alternatives. This argues for experience-adjusted compensation benchmarking rather than flat raises.

3. **Engagement is multidimensional.** The composite EngagementScore (averaging four satisfaction dimensions) is more predictive than any individual satisfaction metric, supporting the use of holistic engagement surveys rather than single-item measures.

4. **Stock options are an effective retention lever.** The 0.53 vs. 0.85 stock option differential between departing and retained employees, combined with StockOptionLevel's high SHAP importance, suggests that equity-based compensation is a particularly effective retention incentive.

5. **Entry-level employees are most vulnerable.** Job Level 1 attrition (26.34%) is 2.7× the rate at Level 2 (9.74%). Early-career employees may benefit from structured mentorship, faster promotion tracks, and competitive starting compensation.

5.2 Practical Implications

The system enables several concrete HR workflows:

- **Targeted retention budgets**: Direct limited retention resources (bonuses, promotions, schedule adjustments) toward employees identified as high-risk rather than distributing them uniformly.
- **Manager alerts**: Automatically notify line managers when a direct report's risk score crosses the medium or high threshold, triggering a structured stay-interview.
- **Scenario planning**: The what-if simulator enables HR to model the impact of proposed interventions (e.g., "if we grant stock options to all Level-1 Sales Reps, how does their risk change?") before committing resources.
- **Department-level strategy**: The department analytics view identifies structural attrition hot spots (e.g., Sales at 20.63%) for targeted policy changes.

5.3 Limitations

1. **Cross-sectional data**: The dataset is a single snapshot without temporal dynamics. Longitudinal data (e.g., satisfaction changes over time) could improve prediction accuracy.

2. **Self-reported satisfaction**: Survey-based satisfaction metrics are subject to response bias and social desirability effects.

3. **Generalizability**: The model is trained on Palo Alto Networks data; applicability to other organizations requires retraining or transfer-learning approaches.

4. **Causal inference**: The model identifies correlational patterns, not causal relationships. Interventions informed by the model should be evaluated via controlled experiments (e.g., A/B testing retention programs).

5. **Ethical considerations**: Predictive attrition systems may raise privacy and fairness concerns. Deployment should include bias auditing across protected attributes (gender, age) and transparent communication with employees.

6 System Deployment

The predictive system is operationalized as a Streamlit web application comprising six interactive dashboard modules:

1. **Executive Dashboard**: Real-time KPI cards (total employees, risk distribution, average attrition probability), risk distribution pie chart, probability histogram, department-level stacked bar chart, and income-vs-risk scatter plot. A sortable table highlights the top 20 highest-risk employees.

2. **Employee Risk Profiles**: Filterable employee table with department, role, and risk-category selectors. Individual deep-dive view shows demographics, compensation, an engagement radar chart (employee vs. company average), and a risk badge.

3. **Department Analytics**: Aggregated department and role risk metrics, overtime impact analysis, and a Job Satisfaction × Work-Life Balance risk heatmap.

4. **Model Performance**: Side-by-side metrics table (Accuracy, Precision, Recall, F1, ROC-AUC, CV Mean/Std), model comparison radar chart, ROC curves, confusion matrix, and expandable hyperparameter panels.

5. **Feature Explainability**: Random Forest importance (top 20), SHAP value analysis (top 15), and Overtime × Satisfaction interaction visualization.

6. **What-If Simulator**: Seventeen adjustable employee attribute controls, real-time risk gauge, rule-based retention recommendations, and a sensitivity analysis chart sweeping seven key features across 10 steps.

Performance optimization: Model training (with hyperparameter search) requires approximately 3–5 minutes on first run. To eliminate this latency on subsequent launches, trained models are persisted to disk using `joblib` with an MD5-based data fingerprint for cache invalidation. Subsequent starts load cached models in under 1 second.

Technology stack: Python 3.x, Streamlit 1.55, scikit-learn 1.8.0, XGBoost 3.2.0, imbalanced-learn 0.14.1, SHAP 0.51.0, Plotly 6.6.0, Pandas 2.3.3, NumPy 2.4.2.

7 Conclusion and Future Work

This paper presents an end-to-end machine learning system for predicting and managing employee attrition at Palo Alto Networks. By combining rigorous model development (four algorithms, hyperparameter tuning, cross-validation, SMOTE) with interpretable explanations (SHAP, feature importance) and a production-ready interactive dashboard, the system transforms attrition prediction from a reactive exercise into a proactive, data-driven capability.

The XGBoost model, achieving the highest ROC-AUC, enables reliable identification of at-risk employees, while the three-tier risk scoring framework translates probabilities into actionable categories. Exploratory analysis reveals that overtime, compensation, stock options, and engagement are the primary levers for retention—findings that directly inform the system's automated retention recommendations.

Future work should pursue the following directions:

- **Longitudinal modeling**: Incorporate temporal features (e.g., satisfaction trends) using recurrent models or survival analysis.
- **Deep learning**: Explore neural network architectures for tabular data (e.g., TabNet, FT-Transformer) for potential accuracy gains.
- **Causal inference**: Apply techniques such as propensity score matching or instrumental variables to isolate causal effects of interventions.
- **Fairness auditing**: Systematically evaluate model predictions for disparate impact across demographic groups.
- **Real-time integration**: Connect the prediction pipeline to live HRIS data feeds for continuous risk monitoring.
- **Cost-sensitive optimization**: Incorporate the differential cost of attrition by role and seniority into the model's objective function.

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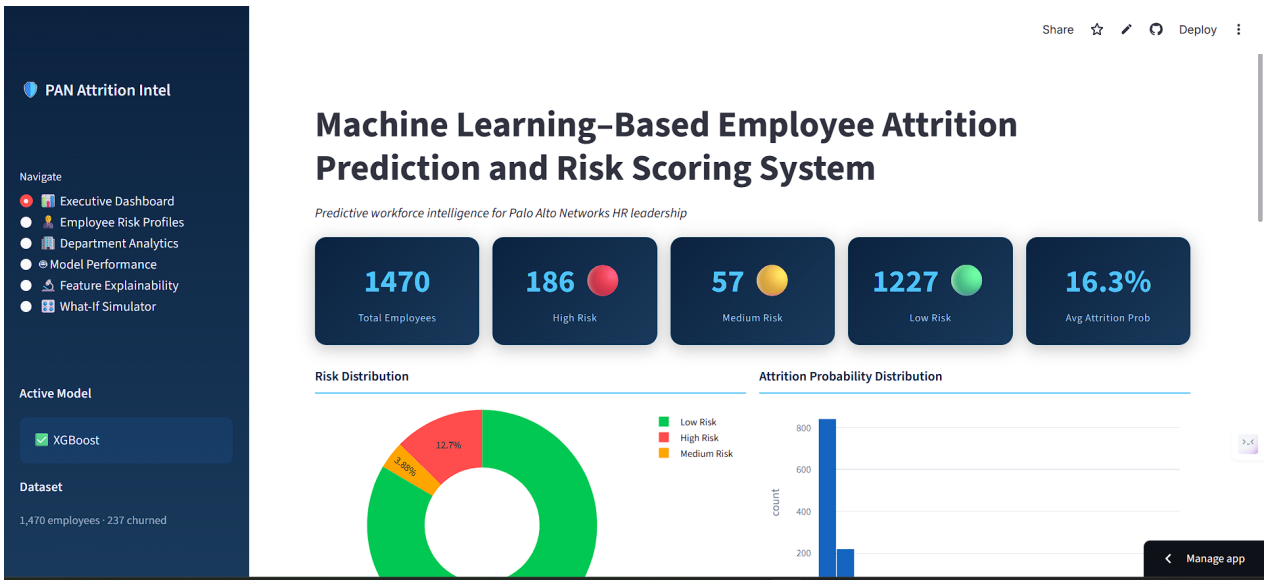

Figure 5: Interactive Streamlit Dashboard for the Machine Learning-Based Employee Attrition Prediction and Risk Scoring System. The dashboard includes executive KPIs, risk distribution visualization, department-level insights, model performance metrics, SHAP explainability, and a what-if simulator for retention scenario planning.

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